

Commutative Algebra

Atiyah - Macdonald

Chap-1

Revision

(i) $\hookrightarrow$ Zorn's lemma $\rightarrow$ $\mathcal{A}$. ideal in $A \Rightarrow \exists$ maximal ideal $\mathfrak{m} \supset \mathcal{A}$.

Pf $\div$ In $A/\mathcal{A}$, applying Zorn's lemma, $\exists$ maximal ideal $\mathfrak{m}'$,

consider the canonical homomorphism $\phi: A \rightarrow A/\mathcal{A}$
then, there is a correspondence b/w $\mathfrak{m}'$ & $\phi^{-1}(\mathfrak{m}')$
& $\phi^{-1}(\mathfrak{m}') \in A$ & is maximal & $\mathcal{A} \in \phi^{-1}(\mathfrak{m}')$
as $\mathcal{A} = \ker \phi$. $\blacksquare$

Cor $\div$ every non-unit in ring R is contained in some maximal ideal.

Ex: (i), A ring, $\mathfrak{m} \subseteq A \ni \forall x \in A - \mathfrak{m}, (x) = (1)$. Then,
 A is local & $\mathfrak{m}$ is maximal.

Pf: suppose $\mathfrak{m}$ is not maximal, then $\exists$ maximal ideal $I \ni$
 $\mathfrak{m} \subsetneq I \subsetneq (1)$, but picking $x \in I \setminus \mathfrak{m}$, as $(x) = (1)$, this
means $I = (1)$, hence its not maximal, $(\Rightarrow \Leftarrow)$. Thus $\mathfrak{m}$ is maximal.
Suppose $\mathfrak{m}'$ is another maximal ideal. Again,
let $a \in \mathfrak{m}' \setminus \mathfrak{m}$. This means $(a) = (1)$,
but as $a \in \mathfrak{m}'$, thus $\mathfrak{m}' = (1)$, $(\Rightarrow \Leftarrow)$

Thus $\mathfrak{m}$ is the only maximal ideal.

(ii) $A \rightarrow$ ring, $\mathfrak{m} \nsubseteq A \ni \forall x \in 1 + \mathfrak{m}, (x) = (1)$, then A is local.

Pf: Use part (i) $\rightarrow$ let $x \in A - \mathfrak{m}$, this means $(\mathfrak{m}, x) = (1)$,
hence $\exists \alpha_1, \alpha_2 \in A \ni \mathfrak{m} \alpha_1 + \alpha_2 x = (1)$, i.e,
 $\alpha_2 x = 1 + (-\alpha_1) \mathfrak{m}$, hence $\alpha_2 x$ is a unit $\rightarrow x$ is a unit
Part i $\Rightarrow$ (local).

Intuition $\div$ If every x outside an ideal is a unit, the ideal must
be super cool!

Propⁿ: The nilradical of a ring A is $\bigcap_{\mathfrak{p} \text{ is prime}} \mathfrak{p}$.

Proof: Let $\mathcal{N}$ be $\text{nil}(A)$. Then, $\mathcal{N} \subseteq A$ & let $x \in \mathcal{N}$. Let $\mathfrak{p}$ be an arbitrary prime ideal in A .

Claim: $x \in \mathfrak{p}$.

Pf: $\exists k \in \mathbb{Z}^+$ with $x^k = 0$, & $x \cdot x^{k-1} = 0$. As $0 \in \mathfrak{p}$, hence either $x \in \mathfrak{p}$ or $x^{k-1} \in \mathfrak{p}$. If $x \in \mathfrak{p}$, done, else, $x^{k-1} \in \mathfrak{p}$, & repeat $\rightarrow$ to eventually have $x^2 \in \mathfrak{p}$, hence $x \in \mathfrak{p}$ is forced.

Using the claim, as $x \in \mathfrak{p}$ for any arbitrary prime ideal $\mathfrak{p}$, we have $x \in \bigcap_{\mathfrak{p} \text{ prime}} \mathfrak{p}$, thus $\mathcal{N} \subseteq \bigcap_{\mathfrak{p} \text{ prime}} \mathfrak{p}$.

For the reverse inclusion, suppose x is not nilpotent.

hence, $\{1, x, x^2, \dots\} \not\subseteq 0$. key $\rightarrow$ Construct a 'bad' prime ideal $\rightarrow$ Zorn's lemma $(1, x, x^2, \dots) \not\subseteq \mathfrak{p}$

Zorn's lemma:

let $\mathcal{A}$ be the collection of all ideals $\mathfrak{a}$

$\ni \forall n > 0, x^n \notin \mathfrak{a}$.

$(\bigcup_{\alpha} \mathfrak{a}_{\alpha})$

Then, by inclusion order, as every chain has an upper bound, $\mathcal{A}$ has a maximal element $\mathfrak{m}'$.

Claim: $\mathfrak{m}'$ is prime

Proof: suppose not, then $\exists a, b \notin \mathfrak{m}'$ but $ab \in \mathfrak{m}'$.

consider $(a, \mathfrak{m}')$ & $(b, \mathfrak{m}')$. As $\mathfrak{m}$ was maximal, hence

$$\exists k_1, k_2 \in \mathbb{Z}^+ \ni \left. \begin{array}{l} a\alpha_1 + \mathfrak{m}'\alpha_2 = x^{k_1} \\ b\beta_1 + \mathfrak{m}'\beta_2 = x^{k_2} \end{array} \right\} \alpha_1, \alpha_2, \beta_1, \beta_2 \in A$$

(Key Motivation: maximality exploit, & multiplying the representations $\rightarrow$ as $a, b \notin \mathfrak{m}'$ while $ab \in \mathfrak{m}'$!)

as $ab \in \mathfrak{p}$ but $a, b \notin \mathfrak{p}$, this is standard

$$\begin{aligned} \therefore x^{k_1+k_2} &= (ab)\alpha_1\beta_1 + am'\alpha_1\beta_2 + bm'\alpha_2\beta_1 + m'^2\alpha_2\beta_2 \\ &= (ab)\alpha_1\beta_1 + (\alpha_1\beta_2 + b\alpha_2\beta_1)m' + m'^2\alpha_2\beta_2 \\ &\& \text{ hence, } x^{k_1+k_2} \in (ab, m') = m' \text{ as } ab \in m', \\ &\text{but this is a contradiction! Hence, } m' \text{ is prime.} \end{aligned}$$

Thus, $\exists$ prime ideal $\mathfrak{p} \not\supseteq \mathfrak{m}$, hence, $x \notin \bigcap_{\mathfrak{p} \text{ prime}} \mathfrak{p}$.

thus, $x \notin \text{Nil}(A) \Rightarrow x \notin \bigcap_{\mathfrak{p} \text{ prime}} \mathfrak{p}$. done. (Contraposition)

x is NOT nilpotent $\Rightarrow x \notin \bigcap \mathfrak{p}$

(Translation: If x is not nilpotent, there exists at least one prime ideal $\mathfrak{p}$ that does not contain x .)

If we can find just **one** prime ideal that avoids x , then x cannot be in the intersection of *all* primes. The proof is done.

2. Why Zorn's Lemma? (The "Growing" Strategy)

We need to find a prime ideal $\mathfrak{p}$ such that $x \notin \mathfrak{p}$. Actually, we need strictly more: if x is not nilpotent, then no power of x is zero. We want a prime ideal that avoids the entire set of powers $S = \{1, x, x^2, x^3, \dots\}$.

The Problem: We don't know what the prime ideals look like in this arbitrary ring. We can't just "pick" one. **The Solution:** We grow one.

1. Start with an ideal that definitely doesn't contain any power of x (e.g., the zero ideal (0) works, assuming x isn't nilpotent).
2. Expand that ideal bigger and bigger, adding as many elements as possible, **but never** allowing any power of x to get inside.
3. **Zorn's Lemma** guarantees that if we keep doing this, we will eventually hit a "ceiling"—a maximal ideal with respect to the property of "missing S ."

(This procedure follows for any multiplicative set rather, $\{1, x, x^2, \dots\}$ wasn't special.)

key: Also motivates that $\sqrt{I} = \{x : x^n \in I \text{ for some } n \in \mathbb{Z}^+\}$

$$= \bigcap_{\substack{\mathfrak{p} \supseteq I \\ \mathfrak{p} \text{ prime}}} \mathfrak{p}, \text{ see Spec}(R) \& \text{ return.}$$

(Prime Avoidance Lemma)

Propⁿ: Let $\mathfrak{p}_1, \mathfrak{p}_2, \dots, \mathfrak{p}_n$ be prime ideals in A ,
(i) & let $\mathfrak{a} \subseteq \bigcup_{1 \leq i \leq n} \mathfrak{p}_i$ be an ideal. Then

$\mathfrak{a} \subseteq \mathfrak{p}_i$ for some $1 \leq i \leq n$.

Proof:- Induct on n $\div$ trivially holds when $n=1$.

suppose holds for $n=k$, i.e.

$$d \subseteq \bigcup_{1 \leq i \leq k} p_i \Rightarrow d \subseteq p_i, \text{ for some } 1 \leq i \leq k.$$

for $n=k+1$, $d \subseteq \bigcup_{1 \leq i \leq k+1} p_i$, rearranging indices if needed

if $d \subseteq \bigcup_{1 \leq i \leq k} p_i$ for some indexing, we are done

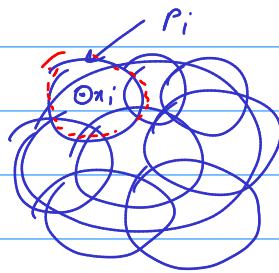
by induction hypothesis. Else, for any indexing,

$$\text{we have } d \not\subseteq \bigcup_{1 \leq i \leq k} p_i \text{ but } d \subseteq \bigcup_{1 \leq i \leq k+1} p_i.$$

This means $d \cap p_i \neq \emptyset \forall 1 \leq i \leq k+1$.

suppose that $d \not\subseteq p_i \forall 1 \leq i \leq k+1$.

for each i , pick $x_i \in d$ but $x_i \notin p_j \forall j \neq i, 1 \leq i \leq k+1$
 thus $x_i \notin \bigcup_{\substack{1 \leq j \leq k+1 \\ j \neq i}} p_j$.



But then, letting
 $a_i = x_i + \prod_{\substack{1 \leq j \leq k+1 \\ j \neq i}} x_j$

as p_j 's are all prime ideals, hence $\prod_{\substack{1 \leq j \leq k+1 \\ j \neq i}} x_j \notin p_j$ (as $x_i \notin p_j \forall j \neq i$)

for any $1 \leq j \leq k+1, j \neq i$, hence $\prod_{\substack{1 \leq j \leq k+1 \\ j \neq i}} x_j \notin \bigcup_{\substack{1 \leq j \leq k+1 \\ j \neq i}} p_j$

Thus, $a_i \notin p_l \forall 1 \leq l \leq k+1$ as

for $l=i$, $x_i \in p_i$ but $\prod_{\substack{1 \leq j \leq k+1 \\ j \neq i}} x_j \notin \bigcup_{\substack{1 \leq j \leq k+1 \\ j \neq i}} p_j$, hence $a_i \notin p_i$

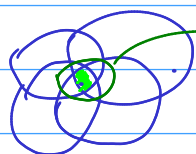
for $l \neq i$, it is by construction not in p_l .

But $a_i \in d$ from RHS while $a_i \notin d$ from the fact that $d \subseteq \bigcup_{1 \leq j \leq k+1} p_j$ & $a_i \notin \text{RHS} \forall i$. contradiction.

(Dual)(ii) Let $\mathfrak{a}_1, \mathfrak{a}_2, \dots, \mathfrak{a}_n$ be ideals in A & $\mathfrak{p}$ be a prime ideal containing $\bigcap_{1 \leq i \leq n} \mathfrak{a}_i$. Then, $\mathfrak{p} \supseteq \mathfrak{a}_j$ for some $1 \leq j \leq n$.

(If $\mathfrak{p} = \bigcap_{1 \leq i \leq n} \mathfrak{a}_i$, then $\mathfrak{p} = \mathfrak{a}_j$ for some $1 \leq j \leq n$).

Pf)

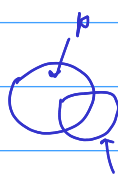


prime $\mathfrak{p}$
containing

Induct

$$\mathfrak{p} \supseteq \bigcap_i \mathfrak{a}_i$$

Suppose $\mathfrak{p} \not\supseteq \mathfrak{a}_i \quad \forall (1 \leq i \leq n)$



Then, $\exists x_i \in \mathfrak{a}_i \ni x_i \notin \mathfrak{p} \quad \forall 1 \leq i \leq n$

Then, $\prod_{1 \leq i \leq n} x_i \notin \mathfrak{p}$, & as $\prod_{1 \leq i \leq n} x_i \in \bigcap_{1 \leq j \leq n} (\mathfrak{a}_j) \subseteq \bigcap_{1 \leq j \leq n} \mathfrak{a}_j$,

but, $\mathfrak{p} \supseteq \bigcap_j \mathfrak{a}_j \rightarrow$ while $\prod_j x_i \subseteq \bigcap_{1 \leq j \leq n} \mathfrak{a}_j$

hence $x_i \in \mathfrak{p}$, a contradiction.

Summary

- Revision of ALG-III ✓
- Zorn's lemma in Nilradical $(= \bigcap_{\mathfrak{p} \text{ prime}} \mathfrak{p})$ ✓
- Local Rings, Criteria ✓

(i) $\mathfrak{m} \leq A \ni \forall x \in A - \mathfrak{m}, (x) = 1 \Rightarrow \mathfrak{m}$ maximal & A local

(ii) $\mathfrak{m}$ maximal in A , then if $\forall x \in 1 + \mathfrak{m}, (x) = 1$, A is local.

- Prime Avoidance Lemma & its dual.

$(\bigcup \mathfrak{p}_i \supseteq \mathfrak{a} \Rightarrow \mathfrak{p}_j \supseteq \mathfrak{a}, \& \bigcap \mathfrak{a}_i \subseteq \mathfrak{p} \Rightarrow \mathfrak{a}_j \subseteq \mathfrak{p})$

if $\bigcap \mathfrak{a}_i = \mathfrak{p}$ then $\mathfrak{a}_j = \mathfrak{p}$

If $\bigcap_{1 \leq i \leq n} \mathfrak{a}_i = \mathfrak{p}$, then $\exists 1 \leq j \leq n \ni \mathfrak{a}_j = \mathfrak{p}$.

Pf:-

Suppose not, then, $\forall 1 \leq j \leq n, \mathfrak{a}_j \neq \mathfrak{p}$.

So, $\exists x_j \in \mathfrak{a}_j \ni x_j \notin \mathfrak{p}$. Hence, $x = \prod_{1 \leq j \leq n} x_j \notin \mathfrak{p}$,

But as $x \in \prod_j (x_j) \subseteq \prod_j \mathfrak{a}_j \subseteq \bigcap_j \mathfrak{a}_j = \mathfrak{p}$,
this is a contradiction.

Hence, $\exists 1 \leq j \leq n \ni \mathfrak{a}_j = \mathfrak{p}$.

(see later - Zariski Topology $\rightarrow$ Karnataka Notes)

A-M Pg. 8 $\rightarrow$ Ex 1.12 $\rightarrow$ (Ideal Quotients, Annihilator)

(i) $(\mathfrak{a}, b) \cong \mathfrak{a} \rightarrow$ trivial

(ii) $((\mathfrak{a}:b):c) = (\{x \in A : x b \subseteq \mathfrak{a}\}:c)$

$$\downarrow$$

$$(\mathfrak{a}:bc) = \{x \in A : x bc \subseteq \mathfrak{a}\}$$

Let $x \in ((\mathfrak{a}:b):c)$. Thus, $x c \subseteq \{x \in A : x b \subseteq \mathfrak{a}\}$.

$\therefore$ Thus, $(x c) b \subseteq \mathfrak{a} \Leftrightarrow x \cdot (bc) \subseteq \mathfrak{a}$

Thus, $((\mathfrak{a}:b):c) \subseteq (\mathfrak{a}:bc)$.

Let $x \in (\mathfrak{a}:bc)$. Then, $x(bc) \subseteq \mathfrak{a}$.

i.e., $(x c) b \subseteq \mathfrak{a} \Rightarrow x c \subseteq (\mathfrak{a}:b) \Rightarrow (\mathfrak{a}:bc) \subseteq ((\mathfrak{a}:b):c)$

Hence the equality.

(ii) $(\mathfrak{a}:b) b \subseteq \mathfrak{a} \rightarrow$ The definition itself!

Exercises

- (7) A ring where $\forall x \in A, x^n = x$ for some $n > 1$, then every prime ideal is maximal.

pf: Let $\mathfrak{p}$ be a prime ideal in A .

Then, $A/\mathfrak{p}$ is an integral domain.

As $A/\mathfrak{p} = \{x + \mathfrak{p} : x \in A\}$, let $\bar{x} \in A/\mathfrak{p}$.

Claim: $A/\mathfrak{p}$ is a field.

Proof: We need to show that every $\bar{x}$ in $A/\mathfrak{p}$ (except possibly $\bar{0}$) has an inverse.

Now, $\bar{x}^n = \bar{x} \Leftrightarrow \bar{x}(\bar{x}^{n-1} - 1) = 0$, as $A/\mathfrak{p}$ is an integral domain, hence either $\bar{x} = 0$ or $\bar{x}^{n-1} = 1$.

If $\bar{x} = 0$, nothing to show.

If $\bar{x}^{n-1} = 1$, this means $\bar{x} \cdot \bar{x}^{n-2} = 1$ ($n \geq 2$)

hence $\bar{x}$ is a unit.

Thus, $A/\mathfrak{p}$ is a field & hence $\mathfrak{p}$ is maximal.

- (10) $A \rightarrow \text{Ring}$, $\text{TFAE} \rightarrow$

(i) A has exactly one prime ideal.

(ii) $\forall x \in A$, either $(x) = (1)$ or $x \in \text{Nil}(A)$.

(iii) $A/\text{Nil}(A)$ is a field ($\Leftrightarrow \text{Nil}(A)$ is maximal)

Proof: $(i \Rightarrow ii) \rightarrow$ Let $\mathfrak{p}$ be the only prime ideal of A .

As $\exists$ $\mathfrak{m}$ maximal ideal & every maximal ideal is prime, hence A must be local, hence $\mathfrak{p}$ is a maximal ideal.

As $\text{Nil}(A) = \bigcap_{\mathfrak{p} \text{ prime}} \mathfrak{p} = \mathfrak{m}$, hence $\forall x \in A$,

either $x \in \mathfrak{m} = \text{Nil}(A)$ or $x \in A - \mathfrak{m}$ (& hence a unit as no other maximal ideal exists), thus $x \in \text{Nil}(A)$ or $(x) = (1)$

((ii) $\Rightarrow$ (iii)) T.S: $\text{Nil}(A)$ is a maximal ideal.

pf: Suppose FTSOC that $\text{Nil}(A)$ is not maximal.

Then, let $x \in A \ni x \notin \text{Nil}(A)$. Then, x is a unit.

But as $\text{Nil}(A)$ is not maximal, hence

$\exists$ $\mathcal{M}$ maximal $\supseteq (\text{Nil}(A), x)$. But as x is a unit, hence $(\text{Nil}(A), x) = (1) \subseteq \mathcal{M}$ which is false.

Thus, $\text{Nil}(A)$ is a maximal ideal.

(iii) $\rightarrow$ (i) : $\text{Nil}(A)$ is maximal. Then it is also prime.

Suppose that there is another prime ideal $\mathfrak{p}$.

Pick $x \in \mathfrak{p} - \text{Nil}(A)$. Then, $(\mathfrak{p}, x)$ is another

ideal. Recalling the lemma in A-M beginning, as

$\text{Nil}(A)$ is maximal & $\forall x \in \text{Nil}(A)$, $1+x$ is a unit,

hence A is local. But $(\mathfrak{p}, x) \subseteq \mathcal{M}$ for some other

maximal ideal $\mathcal{M}$ (as $x \notin \text{Nil}(A)$) & hence, x must

be a unit; which is false as $(\mathfrak{p}) \neq (1)$.

Hence, $\text{Nil}(A)$ is the only prime ideal in A . $\blacksquare$

Much shorter proof with $\text{Spec}(A)$ exists.